Venkata Lakshmi Parimala Pasupuleti

Senior Applied AI Engineer · AI Agents · Business Workflow Automation · RAG · AWS Bedrock · Tool Calling · HITL · Observability · FastAPI · Python

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SUMMARY

Senior applied AI engineer with 9+ years building **production AI agents and business workflow automation** - turning manual processes involving documents, APIs, databases, and enterprise systems into secure, reliable AI-powered solutions. Currently at Intuit building Anthropic Claude integrations into production financial services: **autonomous agent workflows** with tool/function calling, structured outputs, human approvals (HITL), retries, and monitoring; **guardrails** with IAM-aligned least-privilege patterns, input validation, and audit trails; **LLM evaluation harnesses** with CI regression gating. Built **Leasel** - a production agentic platform (LangGraph, 8-agent orchestration) with document-aware RAG (OCR ingestion, vector search, semantic reranking, citation grounding), MCP server integration, HITL approval workflows, and a CI eval harness (93.6% precision, 1,240 test cases). Built **AgentProbe** - an open-source LLMOps platform with cost-aware **AWS Bedrock** routing, CI regression gating, and full per-run observability. At Entrata (7+ years), owned PHI-compliant AWS systems (IAM, S3, least-privilege access) and mission-critical data pipelines under regulated compliance frameworks. Strong full-stack: Python/FastAPI, PostgreSQL, AWS (Lambda, S3, IAM, API Gateway, CloudWatch, SQS, Bedrock), Docker, Kubernetes. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Production AI Agents · Business Workflow Automation · Multi-Step Orchestration | Tool/Function Calling · Structured Outputs · Human Approvals · HITL | Retries · Monitoring · Audit Trails · Production Reliability | RAG · Enterprise Document Search · Vector Search · pgvector · ChromaDB | AWS Bedrock · Lambda · S3 · IAM · API Gateway · SQS · CloudWatch | AI Security · Least-Privilege Access · Guardrails · Credential Management | LLM Evaluation · Observability · CI Regression · LLMOps | Python · FastAPI · REST APIs · Enterprise System Integration | Docker · Kubernetes · CI/CD · GitHub Actions | PostgreSQL · SQL · Async Processing · Event-Driven Systems | LangGraph · LangChain · MCP · Multi-Agent Orchestration | PHI-Compliant Systems · Regulated Data · Compliance Engineering

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA (Remote) — Financial Services AI

- **Production AI agents for business workflow automation:** Build LangChain-based autonomous agent workflows integrating Anthropic Claude - multi-step orchestration with tool/function calling, structured outputs, REST API integrations, and third-party platform connections; human approvals (HITL) for sensitive decisions; retries with exponential backoff; structured audit trails and CloudWatch monitoring for production accountability.
- **Security and guardrails:** Implement IAM-aligned least-privilege access patterns, input validation, sensitive-data handling, and guardrails across AI workflows deployed on AWS; secure credential management via AWS Secrets Manager and parameter store patterns.
- **LLM evaluation and observability:** Build LLM-as-judge eval harnesses with golden datasets and CI regression gates; structured logging per agent run; qualitative error analysis and eval-driven iteration. ML data pipelines (scikit-learn, XGBoost); React/TypeScript front ends; Docker + GitHub Actions CI/CD.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Affordable Housing SaaS (regulated multi-tenant, 100K+ monthly active users)

- Owned PHI-compliant AWS data systems with IAM least-privilege access (Libsodium encryption, S3 migration, 100% coverage, zero incidents); HUD compliance ETL (180K records, zero violations across 6 years); RabbitMQ async event-driven processing with idempotency guarantees and dead-letter routing.
- Led Docker microservices redesign (50% scalability improvement); optimized 200+ SQL queries (30% performance gain); Python/Bash automation eliminating 80% of manual workload; mentored 5+ engineers; 300+ code reviews across 120+ Agile sprints.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built backend modules (Python, PHP, Java, PostgreSQL); delivered large-scale data integration workflows (1.6M records); led AWS S3 migration from legacy storage.

PROJECTS

Leasel - Production AI Agent Platform: Tool Calling · RAG · HITL · Audit Trails

2026 | Python, LangGraph (LangChain), FastAPI, ChromaDB, MCP, AWS, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production AI agent platform automating multi-step document workflows: **8-agent LangGraph orchestration** with tool/function calling, structured outputs, retries, and conditional routing; **MCP server integration** for shared tool access; **enterprise document RAG** (OCR ingestion, ChromaDB vector search, TF-IDF + semantic reranking, citation grounding); **HITL human approval workflows** for regulated decisions with full audit trails; **CI eval harness** (LLM-as-judge, 93.6% precision, 1,240 test cases). Deployed on AWS with structured logging per agent run.

AgentProbe - LLMOps & Observability Platform with AWS Bedrock Routing

2025 | Python, FastAPI, PostgreSQL, AWS Bedrock, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source LLMOps platform: offline golden-dataset evals, LLM-as-judge scoring, CI regression gating, failure handling; **cost-aware model routing including AWS Bedrock** alongside Anthropic, OpenAI, Azure OpenAI, Groq, and Ollama; full per-run tracing and structured logging. Used as a production release gate before every AI deploy.

AI on FHIR — Enterprise API Integration & Secure Identity Controls 2024 I Python, Flask, React, BERT, LLMs, FHIR REST APIs, Docker I github.com/venkatalakshmiparimala/AI_on_FHIR

- Translates natural-language questions into FHIR REST API calls using LLMs, BERT, and embeddings - cut clinical lookup time ~60%. OAuth-secured enterprise API integration, HIPAA-style RBAC with identity controls and least-privilege access, Docker-containerized. Demonstrates secure enterprise system integration patterns.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia I GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017